Q.1 (2 Marks Each)

1. Define Software Metrics. Classify software metrics with one example of each class.
2. What is test suite minimization? State any two benefits of minimizing a test suite.
3. Why is test automation needed? State any four reasons for automating software testing.
4. Differentiate between Static and Dynamic testing tools with one example of each.
5. Differentiate between JIRA and Bugzilla (any four points).
6. List any four challenges in testing web-based software.
7. Differentiate between Traditional Software and Web-based Software (any four points).
8. Define Agile Testing. State any two principles of Agile Testing.
9. List any four quality management principles of ISO 9000:2000.
10. Explain the DMAIC methodology of Six Sigma in brief.

Q.2 (5 Marks Each)

1. A project has the following counts, all of average complexity: External Inputs = 24, External Outputs = 16, External Inquiries = 22, Internal Logical Files = 4 and External Interface Files = 2. The sum of the complexity adjustment factors (ΣFi) is 42. Apply Function Point Analysis to compute the Unadjusted Function Points and the Adjusted Function Points, and explain how function points help in estimating testing effort. ****
2. Explain test suite prioritization, its types and techniques. How is the effectiveness of a prioritized test suite measured (APFD)?
3. A company with frequent regression cycles plans to purchase an automated testing tool. Analyze the guidelines and criteria for selecting a testing tool and the various costs incurred in testing tools, and evaluate whether investing in automation is justified for this project.
4. Apply Selenium WebDriver to automate the login functionality of a web application. Design positive and negative test cases, outline the automation steps (locating elements, entering data, verifying the result) and state the Selenium suite components used.
5. An online shopping website is about to be released. Apply web-based testing techniques to design test cases covering functionality, user interface/usability, browser compatibility, performance (load) and security of the website.
6. Explain the Agile Testing Life Cycle and describe the testing activities carried out in the different phases of Scrum.
7. Analyze McCall's quality factors and criteria for an Internet Banking application. Identify the factors most critical to its success with justification, and evaluate the trade-off (conflict) between any two quality factors.